SRM INSTITUTE OF SCIENCE & TECHNOLOGY
Deemed to be University u/s 3 of UGC Act, 1956

21CSC574J – BIG DATA ANALYTICS USING ARTIFICIAL INTELLIGENCE TECHNOLOGIES

Unit 5

Course Learning Rationale (CLR)

CLR-1 :	Understand the basics of Hadoop and related technologies to big data analytics and Examine the ecosystem tools in hadoop
CLR-2 :	Provide with comprehensive Apache Spark & Spark SQL knowledge for Big data processing
CLR-3 :	Provide with comprehensive in Spark MLlib and structured streaming
CLR-4 :	Recognize the applications with Apache airflow and Zookeeper
CLR-5 :	Explore the advantages of using Ontologies for enhanced data integration and knowledge representation in Big Data contexts.

Course Learning Outcomes (CLO):

At the end of this course, learners will be able to:

CO-1:	Apply MapReduce, HDFS and YARN develop big data applications
CO-2:	Provide with comprehensive Apache Spark & Spark SQL knowledge for Big data processing
CO-3:	Identify the importance of Spark MLlib and structured streaming
CO-4:	Identify the importance of Apache Airflow and Zookeeper tool
CO-5:	Utilize Ontologies to enhance data integration, interoperability, and knowledge representation in Big Data environments.

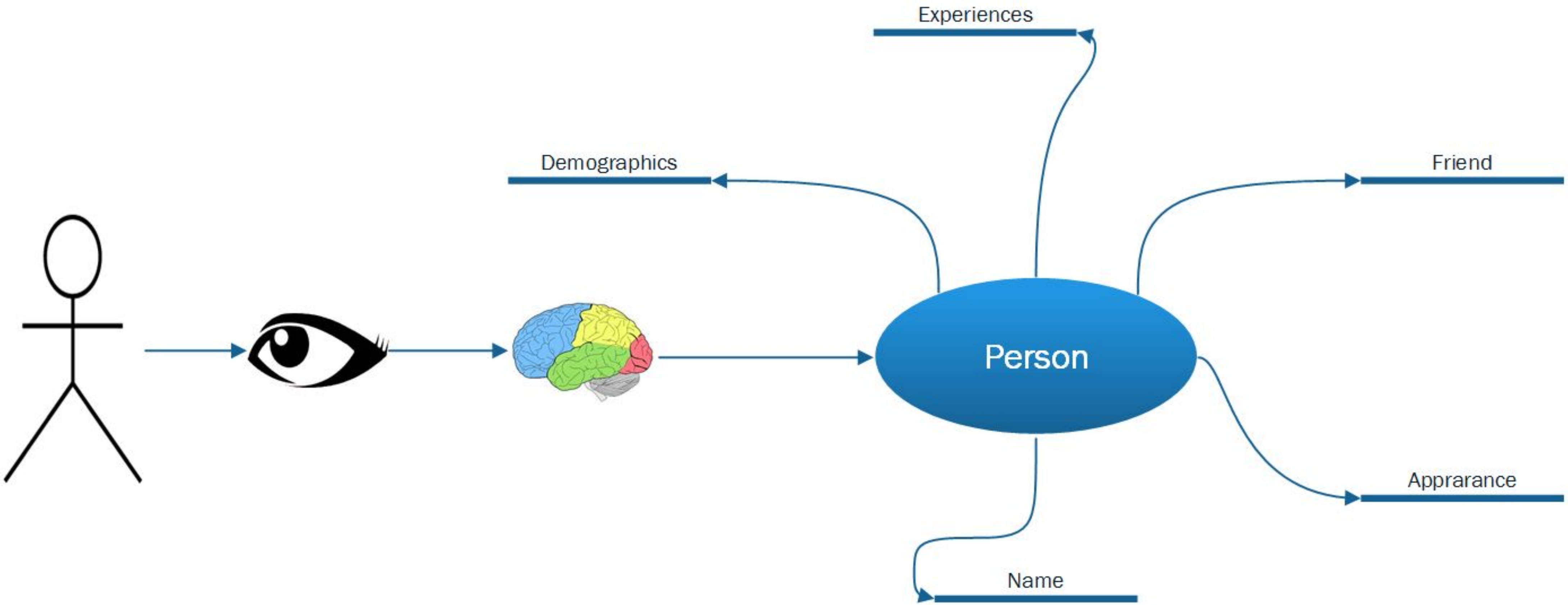
Syllabus

Module -5 Ontology for Big Data

1. *Human brain and Ontology,*
2. *Ontology of information science,*
3. *Ontology properties,*
4. *Advantages of Ontologies,*
5. *Components of Ontologies,*
6. *The role Ontology plays in BigData,*
7. *Ontology alignment,*
8. *Goals of Ontology in big data,*
9. *Challenges with Ontology in Big Data,*
10. *RDF – the universal data format,*
11. *Using OWL, the Web Ontology Language,*
12. *SPARQL query language,*
13. *Building intelligent machines with Ontologies,*
14. *Ontology learning, Ontology learning process.*

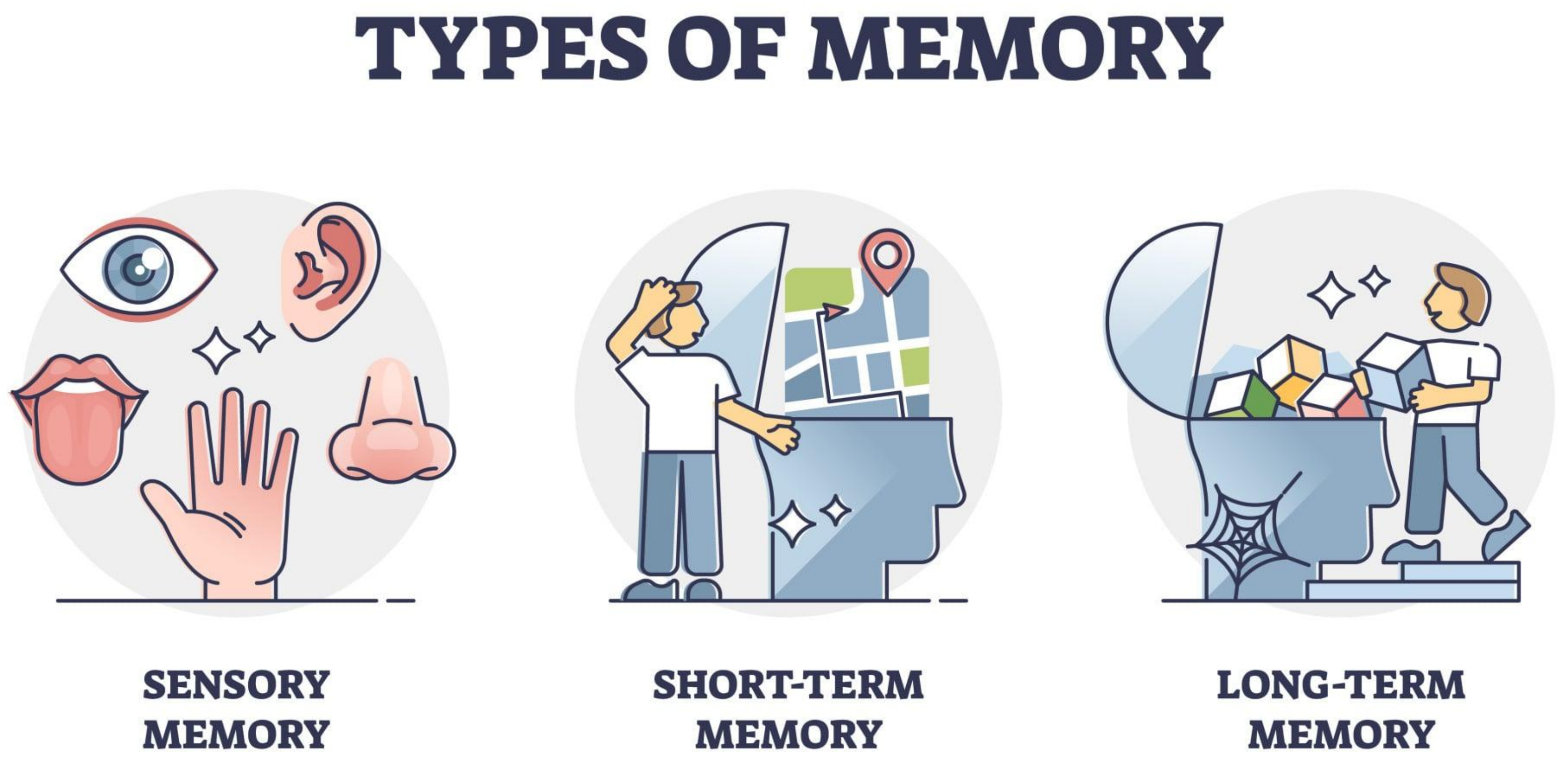
Case studies for Big data using artificial Intelligence Technologies

1. Human brain and Ontology

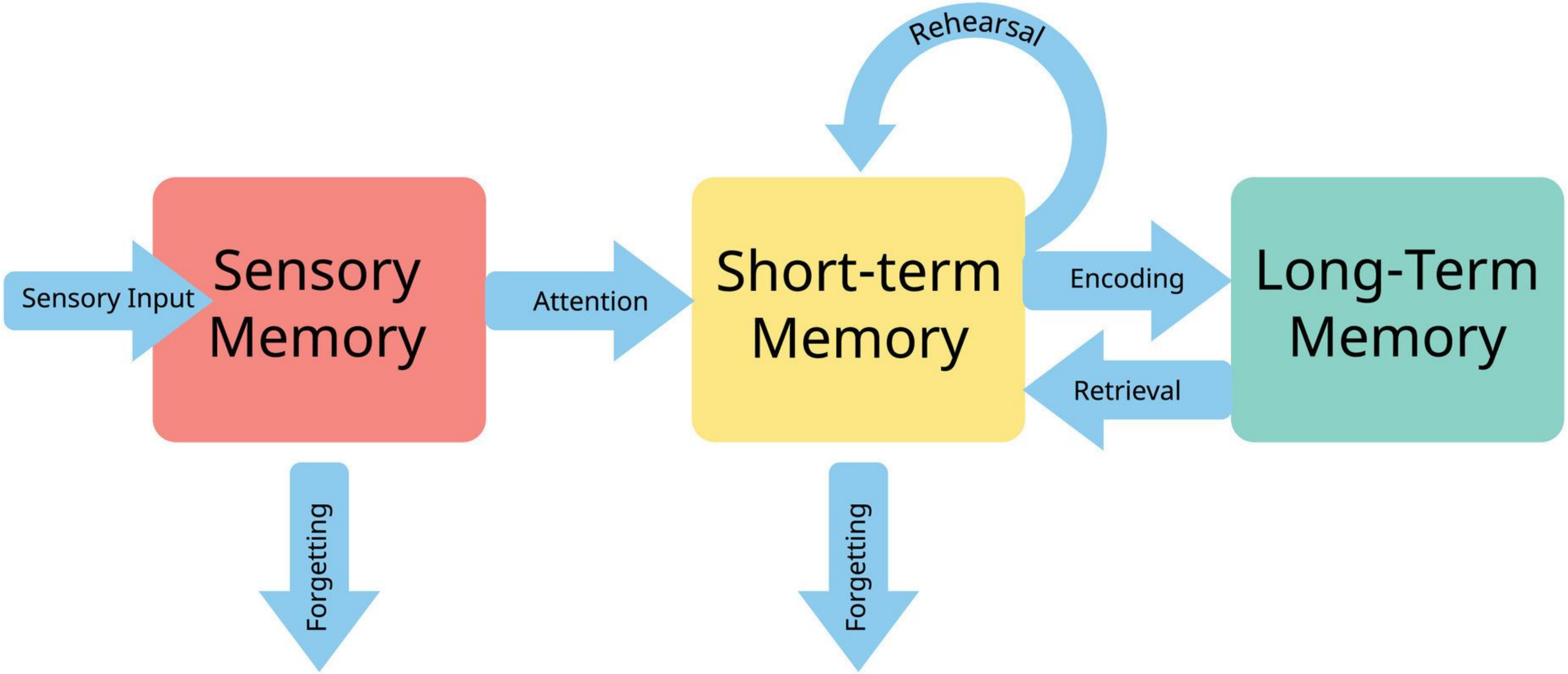


Ontology in brain science provides structured frameworks to classify cognitive processes, brain regions, and cell types, linking theoretical concepts to empirical data.

1. Human brain and Ontology



1. Human brain and Ontology

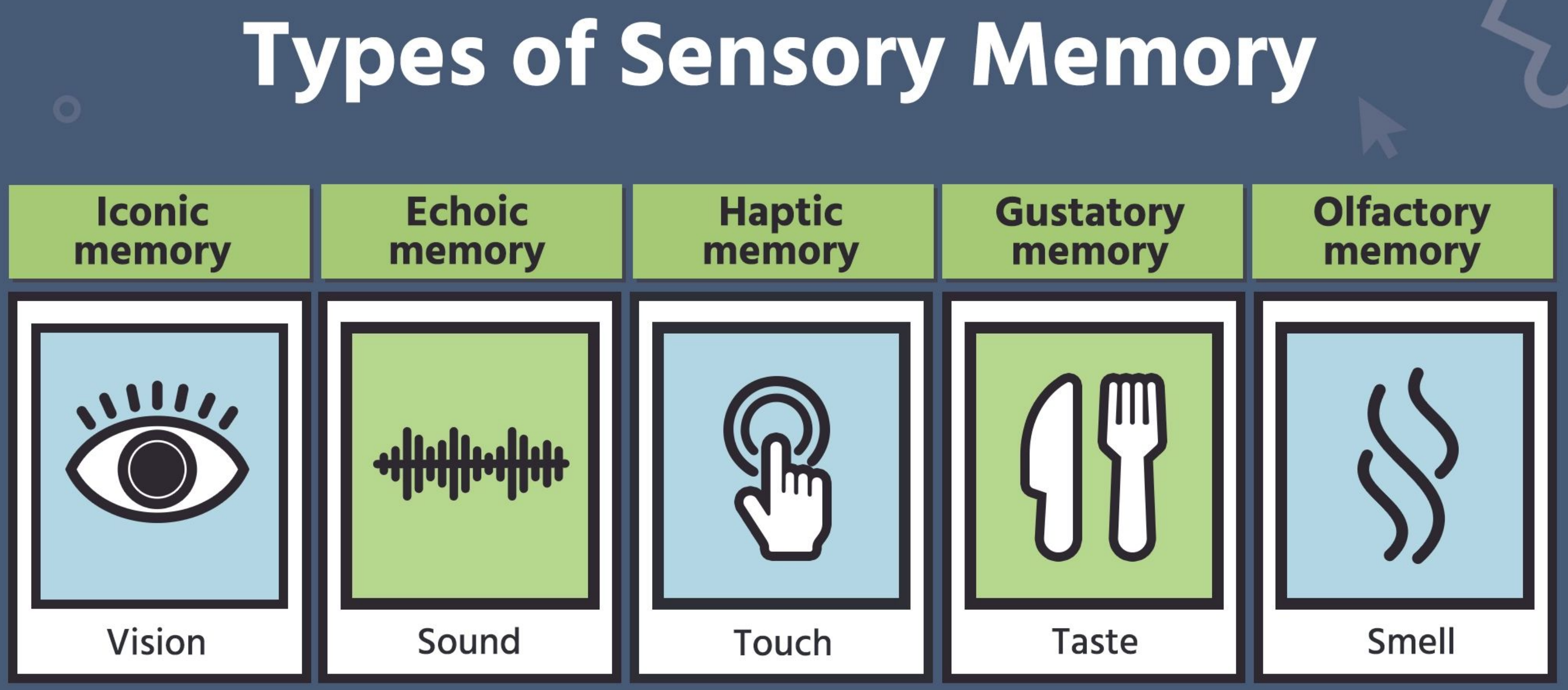


Irrespective of the memory compartment, the information is stored in the form of patterns and links within the human brain. In a memory game that requires players to momentarily

Sensory memory:

This is the first-level memory, and the majority of the information is flushed within milliseconds.

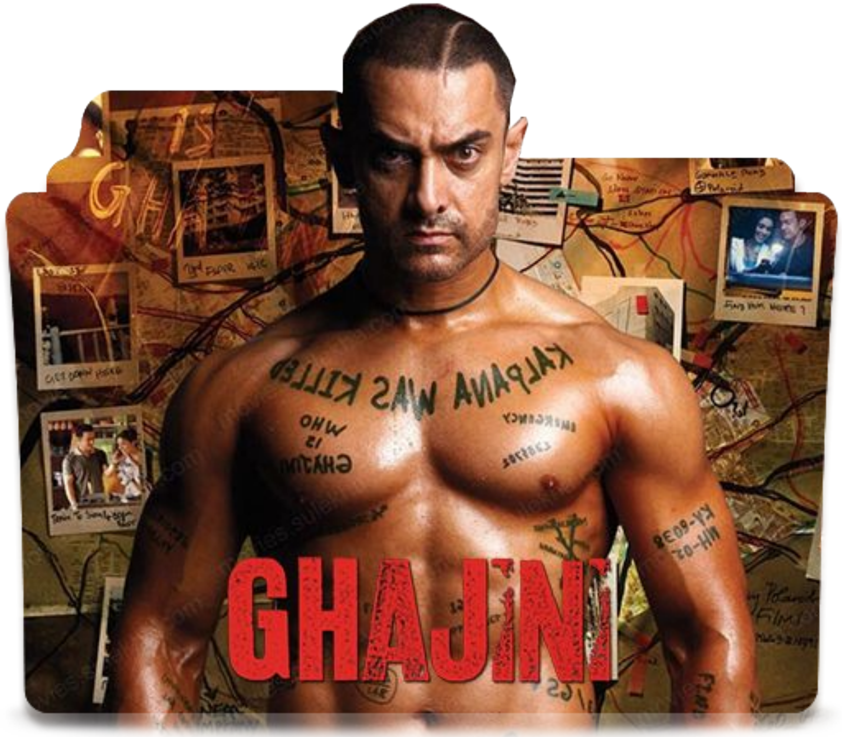
- Consider, for example, when we are driving a car.
- We encounter thousands of objects and sounds on the way, and most of this input is utilized for the function of driving.
- Beyond the frame of reference in time, most of the input is forgotten and never stored in memory.



Short-term memory:

This is used for the information that is essential for serving a temporary purpose.

- Consider, for example, that you receive a call from your coworker to remind you about an urgent meeting in room number D-1482.
- When You start walking from your desk to the room, the number is significant and the human brain keeps the information in short-term memory.
- This information may or may not be stored beyond the context time.
- These memories can potentially convert to long-term memory if encountered within an extreme situation.



Long-term memory:

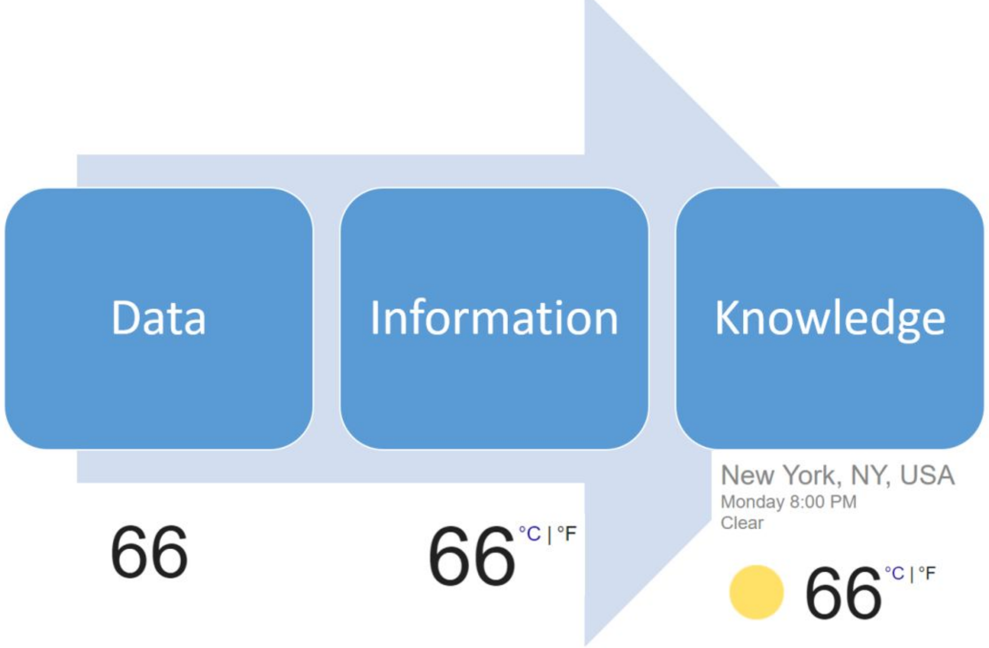
This is the memory that will last for days or a lifetime.

- For example, we remember our name, date of birth, relatives, home location, and so many other things.
- The long-term memory functions on the basis of patterns and links between objects.
- The non-survival skills we learn and master over a period of time, for example playing a musical instrument, require the storage of connecting patterns and the coordination of reflexes within long-term memory.

2. Ontology of information science

Definition:

A formal naming and definition of types, properties, and interrelationships of the entities that fundamentally exist for a particular domain.



3. Ontology properties

At a high level, Ontologies should have the following properties to create a consistent view of the universe of data, information, and knowledge assets:

- The Ontologies should be complete so that all aspects of the entities are covered.
- The Ontologies should be unambiguous in order to avoid misinterpretation by people and software applications.
- The Ontologies should be consistent with the domain knowledge to which they are applicable. For example, Ontologies for medical science should adhere to the formally established terminologies and relationships in medical science.
- The Ontologies should be generic in order to be reused in different contexts.
- The Ontologies should be extensible in order to add new concepts and facilitate adherence to the new concepts, that emerge with growing knowledge in the domain.
- The Ontologies should be machine-readable and interoperable.

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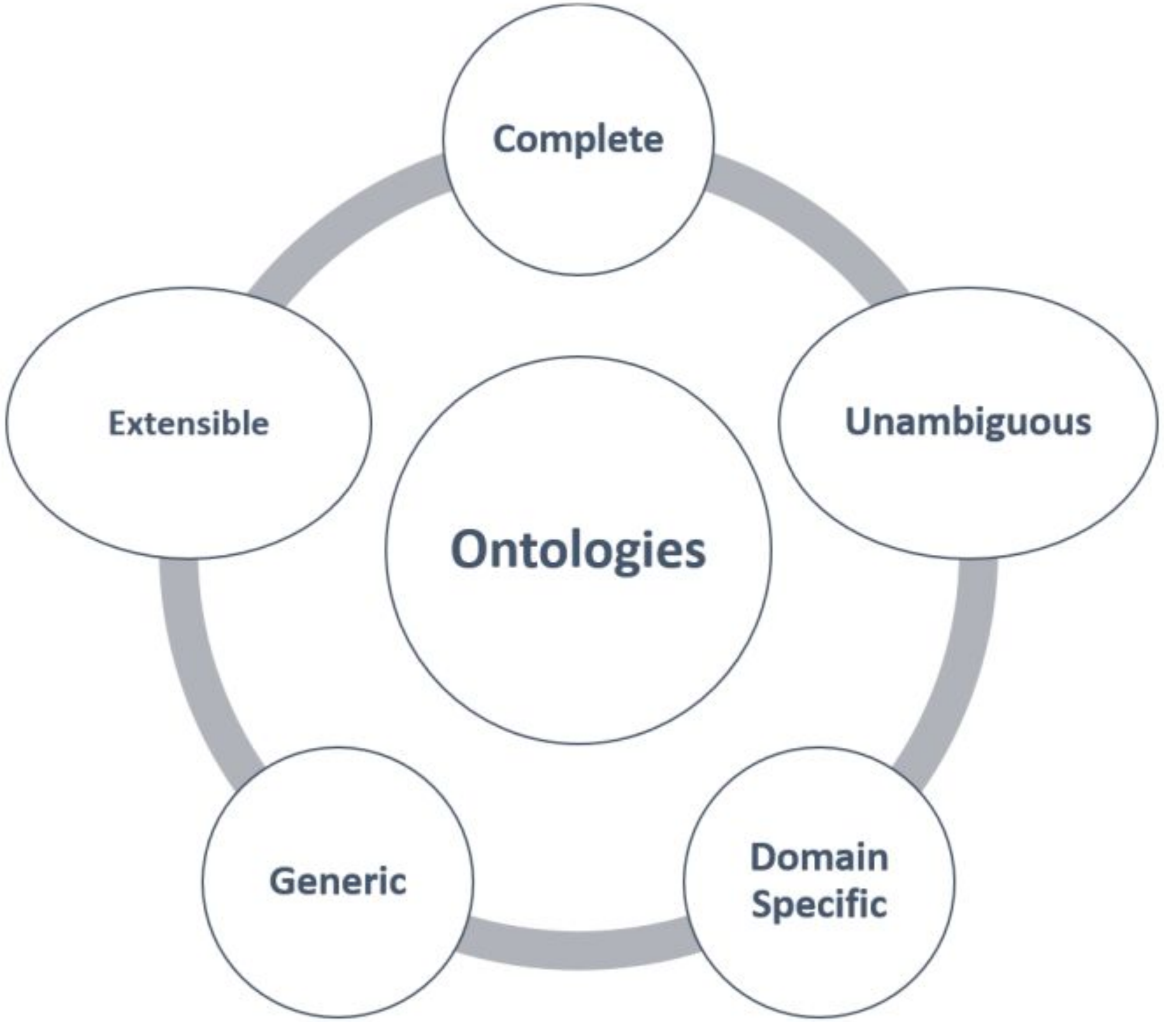

4. Advantages of Ontologies

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The following are the advantages of Ontologies:

- Increased quality of entity analysis
- Increased use, reuse, and maintainability of the information systems
- Facilitation of domain knowledge sharing, with common vocabulary across
- Independent software applications

There are some fundamental differences between databases and Ontologies, as follows:

- Ontologies are semantically richer than the concepts represented by databases
- Information representation in an Ontology is based on semi-structured, natural language text and it is not represented in a tabular format
- The basic premise of Ontological representation is globally consistent terminology to be used for information exchange across domains and organizational boundaries
- More than defining a confined data container, Ontologies focus on generic domain knowledge representation

5. Components of Ontologies

Concepts: These are the general things or entities similar to classes in object-oriented programming, for example, a person, an employee, and so on.

Slots: These are the properties or attributes of the entities, for example, gender, date of birth, location, and so on.

Relationships: These represent interactions between concepts, or is-a, has-a relationships, for example, an employee is a person.

5. Components of Ontologies

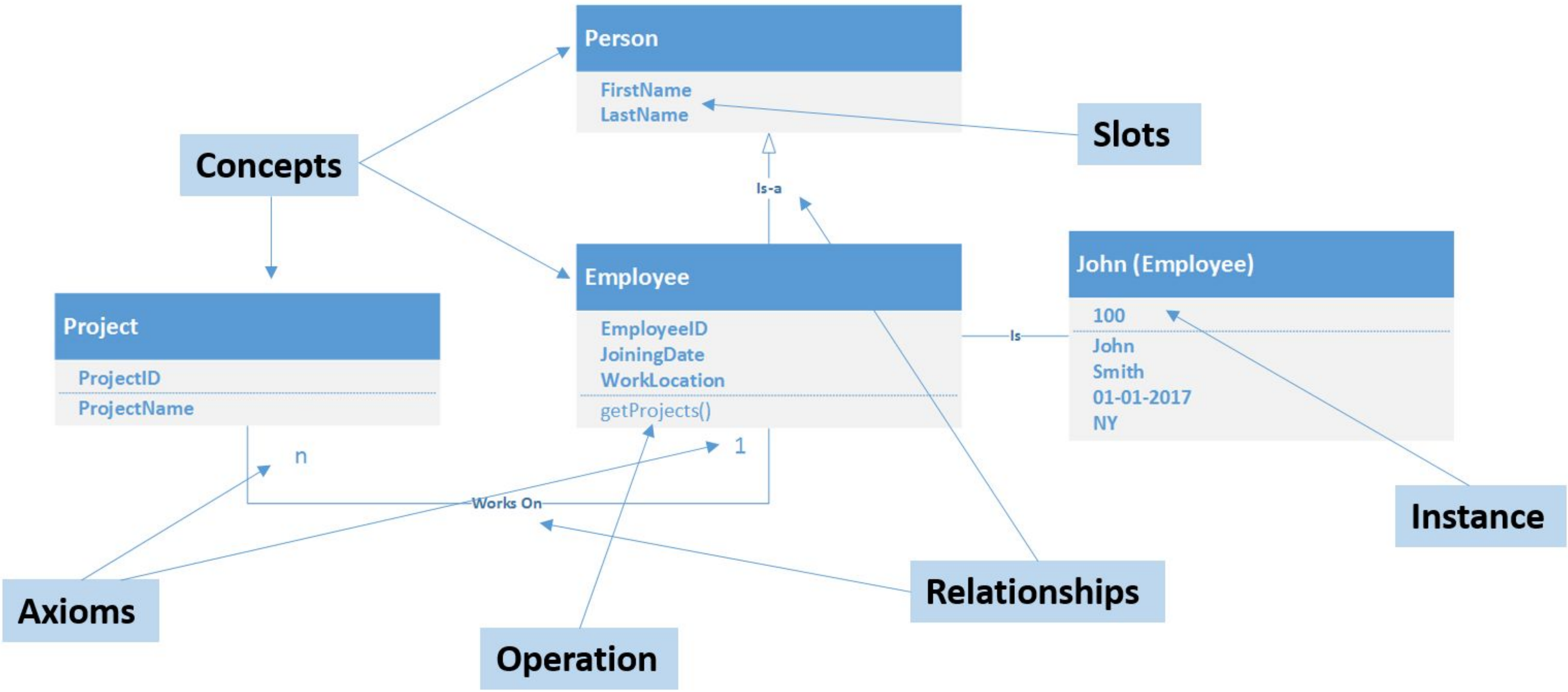
Axioms: These are statements which are always true in regards to concepts, slots and relationships, for example, a person is an employee if he is employed by an employer.

Instances: These are the objects of a class in object-oriented terms.

For example, John is an instance of the Employee class. It is a specific representation of a concept. Ontology, along with instances, fully represents knowledge.

Operations: These are the functions and rules that govern the various components of the Ontologies. In an object-oriented context, these represent methods of a class

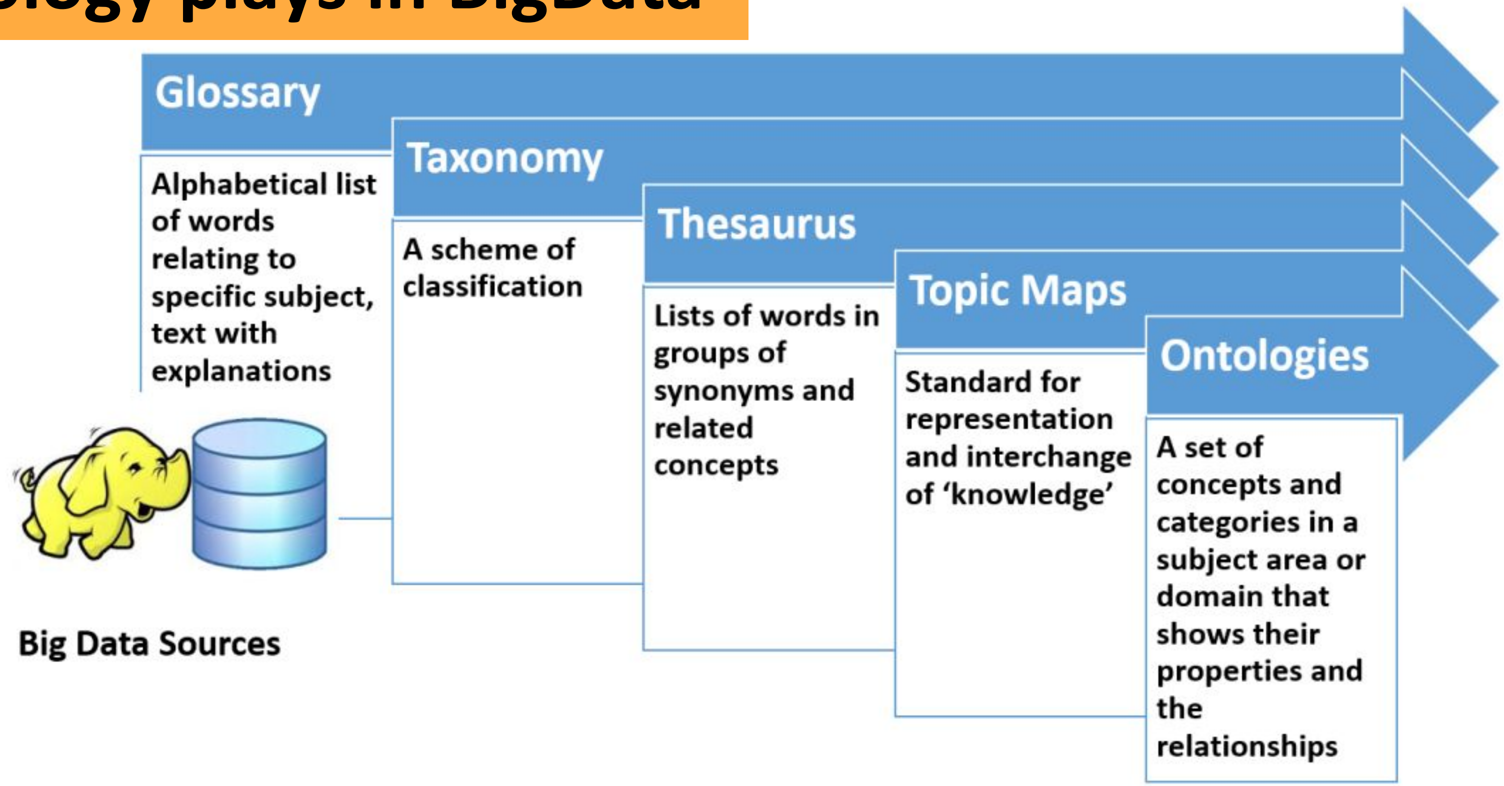
5. Components of Ontologies



6. The role Ontology plays in BigData

- **Big Data Challenge**: Data grows too fast, comes in structured (tables) and unstructured (text, images) formats.
- **Traditional Approach**: ETL (Extract–Transform–Load) requires manual modeling → rigid, slow, hard to change.
- **Ontology Definition**: A way to describe “things” (entities, concepts) and their relationships formally.
- **Why Important**: Helps avoid confusion and false interpretations when dealing with complex, large datasets.
- **Real-Time Example**:
  Google Knowledge Graph → Instead of searching “strings” (keywords), it searches “things” (entities like *Person*, *Place*, *Event*).
  Example: Searching “Jaguar” → Ontology distinguishes between *animal* vs *car brand*.

6. The role Ontology plays in BigData

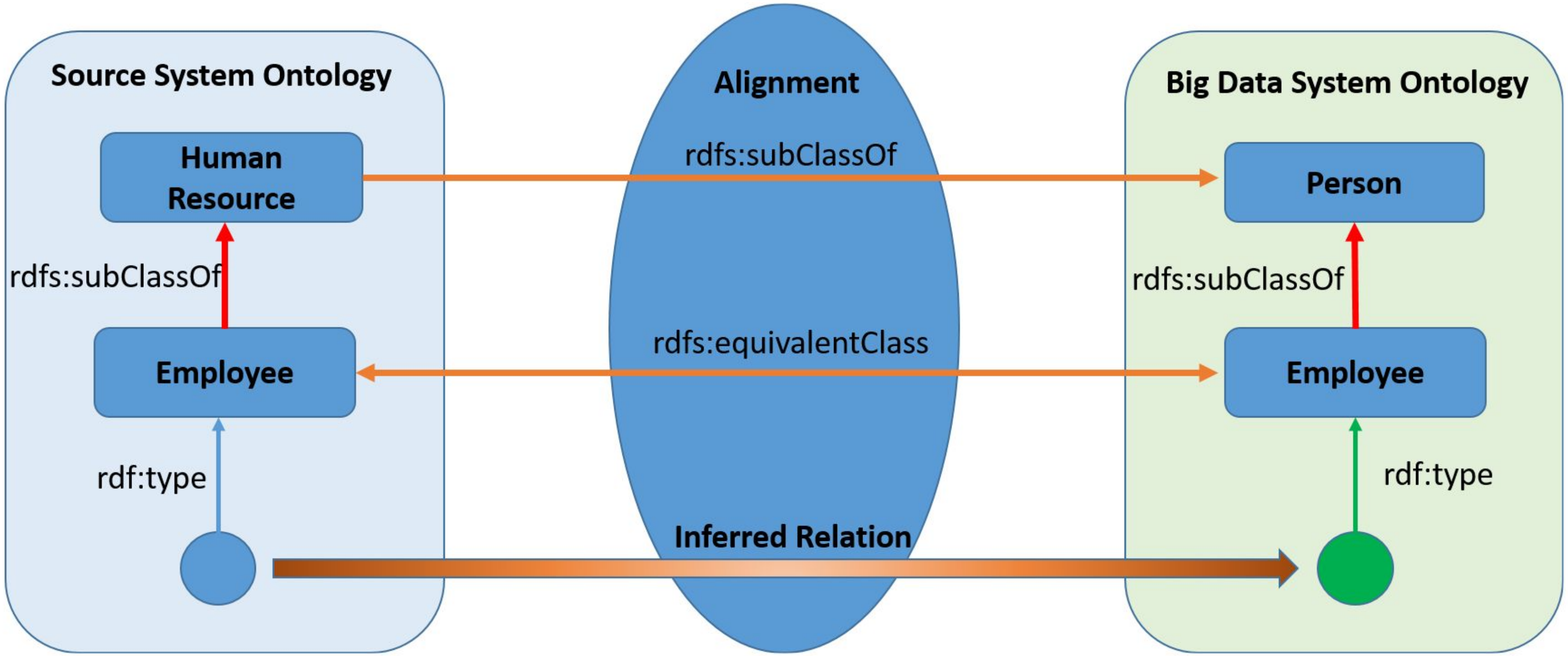


- **Glossary**: At the most basic level, data can be organized into simple word lists with definitions. For example, a glossary of medical terms helps doctors and patients understand terminology.
- **Taxonomy**: Moving further, data can be classified into categories. In e-commerce, products are grouped into categories like *Electronics* → *Mobile Phones* → *Smartphones*.
- **Thesaurus**: Adds relationships between words, such as synonyms and related concepts. In search engines, this helps connect “car” with “automobile” or “vehicle.”
- **Topic Maps**: Provide a standardized way to represent and exchange knowledge. For instance, in education, topic maps can link “Machine Learning” to related subtopics like “Neural Networks” and “Decision Trees.”
- **Ontologies**: The most advanced level, where concepts are formally defined along with their properties and relationships. In Big Data, ontologies enable systems to understand context — e.g., in healthcare, linking “Patient → Disease → Treatment → Outcome” across multiple hospital databases.

7. Ontology alignment

Ontology alignment or matching is a process of determining one-to-one mapping between entities from heterogeneous sources.

Using this mapping, we can infer the entity types and derive meaning from the raw data sources in a consistent and semantic manner:



8. Goals of Ontology in big data

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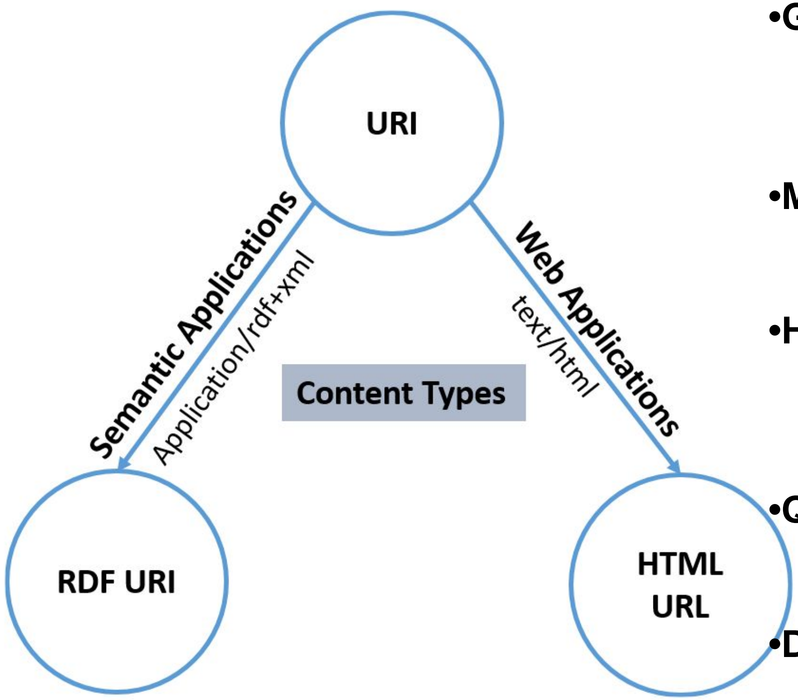
The following are the goals of Ontology in big data:

9. Challenges with Ontology in Big Data

We face the following challenges when using Ontology in big data:



- **Generating Entities (Strings → Things)**
    - Raw data often comes as text (strings).
    - Ontology requires converting these into meaningful “things” (entities like *Person*, *Product*, *Event*).
    - Example: The word “Apple” → Ontology distinguishes between *fruit* and *company*.
- **Managing Relationships**
    - Ontologies define how entities are connected (e.g., *Employee* → *works for* → *Company*).
    - In Big Data, relationships are vast and complex, making them hard to maintain consistently.
- **Handling Context**
    - The same entity can mean different things in different domains.
    - Example: “Jaguar” in wildlife vs. “Jaguar” in automobiles.
    - Ontology must capture this context to avoid misinterpretation.
- **Query Efficiency**
    - Ontology queries (like SPARQL) can be computationally heavy on large datasets.
    - Ensuring fast and efficient retrieval is a major challenge in real-time systems.
- **Data Quality**
    - Ontologies rely on accurate, clean data.
    - In Big Data, sources are heterogeneous (social media, IoT, healthcare), often noisy or incomplete.
    - Poor data quality leads to incorrect mappings and unreliable knowledge.

10. RDF – the universal data format

Resource Description Framework (RDF)

- The **Resource Description Framework (RDF)** is a method for describing and exchanging graph data, originally designed by the World Wide Web Consortium (W3C) as a data model for metadata.
- RDF represents data as a directed graph composed of triple statements, where each statement consists of a subject, predicate, and object.
- These components can be identified by Uniform Resource Identifiers (URIs), and objects can also be literal values.

RDF Structure and Syntax

RDF uses a simple yet powerful data model to represent complex relationships and situations. The basic structure of RDF is a triple, which consists of:

- **Subject**: The resource being described.
- **Predicate**: The property or relationship of the subject.
- **Object**: The value or another resource related to the subject.

For example, the statement "The sky has the color blue" can be represented in RDF as:

- **Subject: "the sky"**
- **Predicate: "has the color"**
- **Object: "blue".**

10. RDF – the universal data format

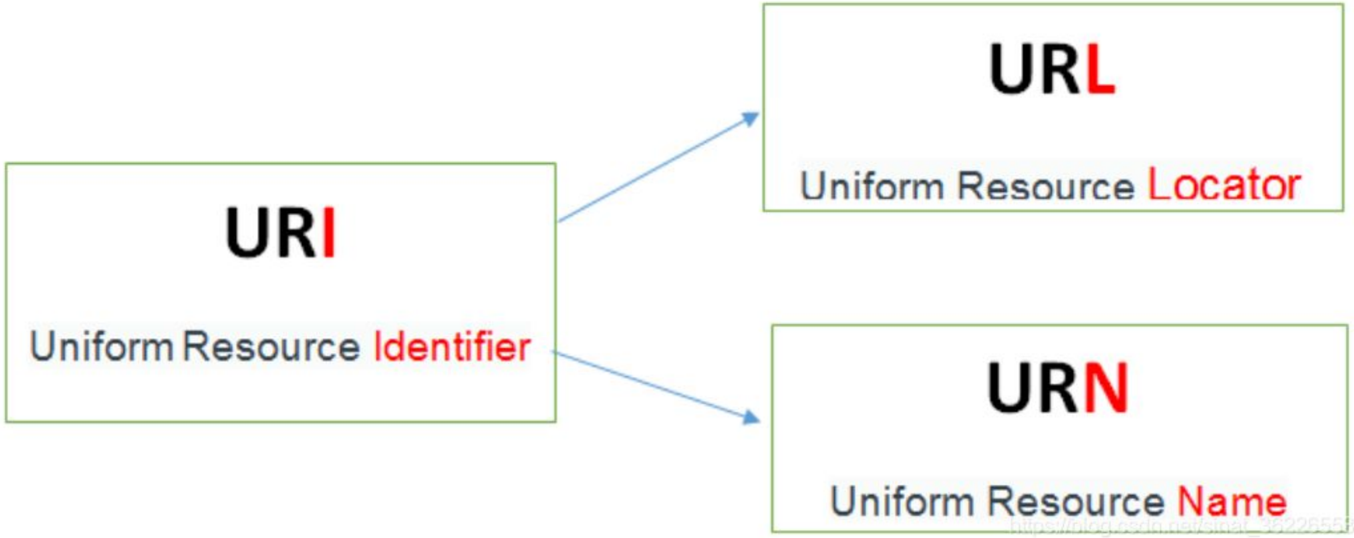
Universal Resource Identifiers (URIs)

URIs (Uniform Resource Identifiers) are the backbone of RDF because they uniquely identify subjects, predicates, and objects without ambiguity.

Unlike plain text labels, URIs ensure that identifiers are globally unique and machine-processable.

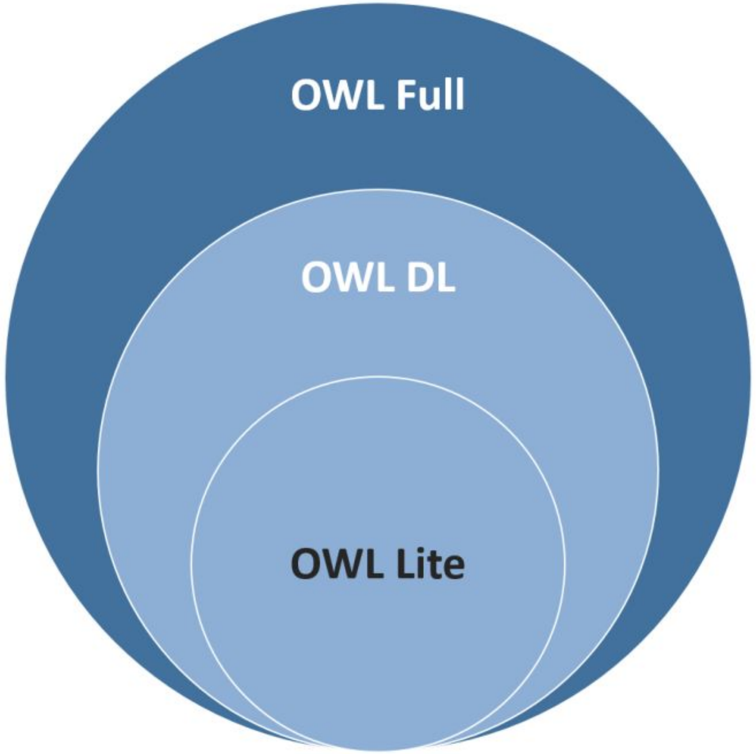
URIs can identify:

- **Network-accessible resources** (e.g., web pages, images).
- **Non-digital entities** (e.g., people, books).
- **Abstract concepts** (e.g., "creator").



Example

```
@prefix eric: <http://www.w3.org/People/EM/contact#>
@prefix contact: <http://www.w3.org/2000/10/swap/pim/contact#>
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
eric:me contact:fullName "Eric Miller"
eric:me contact:mailbox <mailto:e.miller123(at)example>
eric:me contact:personalTitle "Dr."
eric:me rdf:type contact:Person
```



- **OWL DL:** This is used for supporting description logic.
- This supports maximum expressiveness and logical reasoning capabilities.
- This is characterized by:
    - Well-defined semantics
    - Well-understood formal properties for the entities
    - The ease of implementation of known reasoning algorithms
- **OWL Full:** This is based on RDFS-compatible semantics.
- It complements the predefined RDF and OWL vocabulary.
- However, with OWL Full, the software cannot completely reason and inference.
- **OWL Lite:** This is used for expressing taxonomy and simple constraints such as zero-to-one cardinality.

12. SPARQL query language

SPARQL (SPARQL Protocol and RDF Query Language) is a powerful query language and protocol used for querying and manipulating RDF (Resource Description Framework) data. It enables users to retrieve and manipulate data stored in RDF format, which is commonly used for representing semantic web data.

Key Features of SPARQL

SPARQL allows querying across diverse data sources, whether the data is stored natively as RDF or viewed as RDF via middleware. It supports:

- **Basic Graph Patterns**: Matching RDF triples with variables.
- **Optional Patterns**: Including optional data in results without rejecting solutions.
- **Union and Negation**: Combining patterns or excluding specific matches.
- **Aggregations**: Functions like SUM, COUNT, AVG, MIN, and MAX.
- **Subqueries**: Embedding queries within other queries for complex data retrieval.
- **Property Paths**: Querying paths of arbitrary lengths between nodes.
- **Result Formats**: Returning results as tabular data or RDF graphs.

12. SPARQL query language

Example of a Basic Query

Here’s a simple SPARQL query to retrieve the title of a book:

Data:

<http://example.org/book/book1> <http://purl.org/dc/elements/1.1/title> "SPARQL Tutorial«

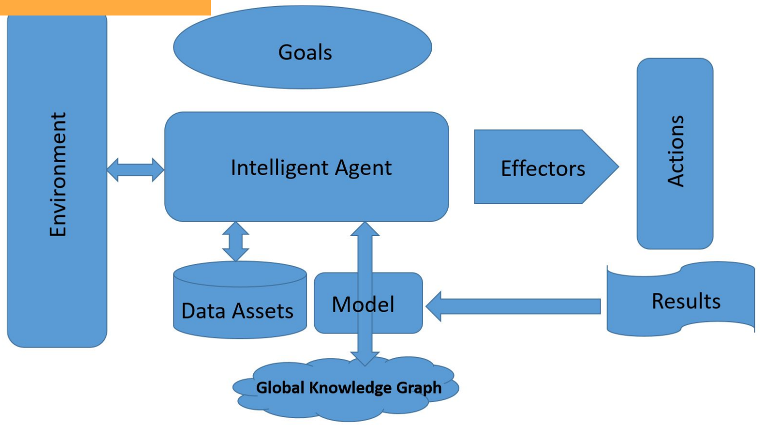
Query:

```
SELECT ?title
WHERE {
 <http://example.org/book/book1> <http://purl.org/dc/elements/1.1/title> ?title
}
```

Result:

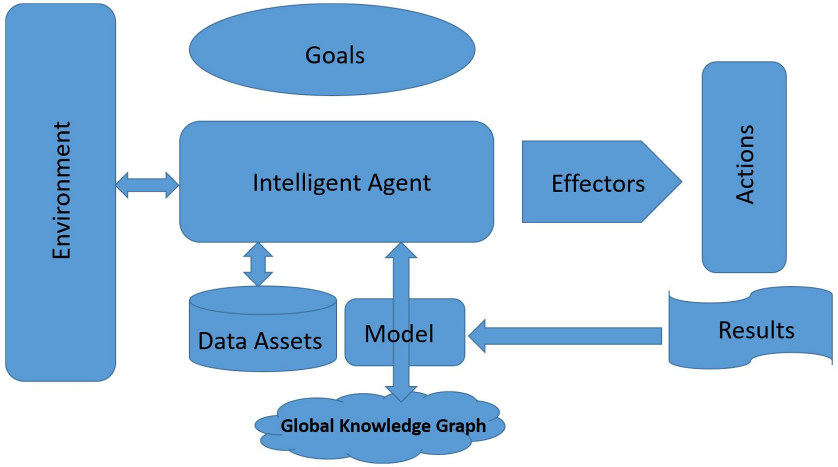
title
"SPARQL Tutorial"

14. Building intelligent machines with Ontologies



- **Goals: Every intelligent system should have a well-defined set of goals.** These goals govern the rational decisions taken by the intelligent system and drive actions and hence results. For example, in the case of an intelligent agent that is responsible for the flight boarding process, one of the goals is to restrict access to anyone who does not pass all security checks, even if the person has a valid air ticket. In defining the goals for intelligent agents, one of the prime considerations should be that the AI agent or systems should complement and augment human capabilities.
- **Environment:** The intelligent agent should operate within the context of the environment. Its decisions and actions cannot be independent of the context. In our example use case, the environment is the airport, the passenger gates, flight schedules, and so on. The agents perceive the environment with various sensors, for example video cameras.
- **Data Assets:** The intelligent agent needs access to historical data in terms of the domain and the context in which it operates. The data assets can be available locally and globally (internet endpoints). These data assets ideally should be defined as RDF schema structures with standardized representations and protocols. These data assets should be queryable with standard languages and protocols (SPARQL) in order to ensure maximum interoperability.

14. Building intelligent machines with Ontologies



- **Model:** This is where the real intelligence of the agent is available as algorithms and learning systems. These models evolve continuously based on the context, historical decisions, actions, and results. As a general rule, the model should perform better (more accurately) over a period of time for similar contextual inputs.
- **Effectors:** These are the tangible aspects of the agent which facilitate actions. In the example of an airline passenger boarding agent, the effector can be an automated gate opening system which opens a gate once all the passengers are fully validated (having a valid ticket, identity, and no security check failures). The external world perceives the intelligent agent through effectors.
- **Actions and Results:** Based on the environmental context, the data assets, and the trained models, the intelligent agent makes decisions that trigger actions through the effectors. These actions provide results based on the rationality of the decision and accuracy of the trained model. The results are once again fed into model training in order to improve accuracy over a period of time.

15. Ontology learning, Ontology learning process.

There are various approaches to Ontology learning, as follows:

- **Ontology learning from text:** In this approach, the textual data is extracted from various sources in an automated manner, and keywords are extracted and classified based on their occurrence, word sequencing, and patterns.
- **Linked data mining:** In this processes, the links are identified in the published RDF graphs in order to derive Ontologies based on implicit reasoning.
- **Concept learning from OWL:** In this approach, existing domain-specific Ontologies are leveraged for expand the new domains using an algorithmic approach.
- **Crowdsourcing:** This approach combines automated Ontology extraction and discovery based on textual analysis and collaboration with domain experts to define new Ontologies. This approach works great since it combines the processing power and algorithmic approaches of machines and the domain expertise of people. This results in improved speed and accuracy.

15. Ontology learning, Ontology learning process.

Here are some of the challenges of Ontology learning:

Dealing with heterogeneous data sources: The data sources on the internet, and within application stores, differ in their forms and representations. Ontology learning faces the challenge of knowledge extraction and consistent meaning extraction due to the heterogeneous nature of the data sources.

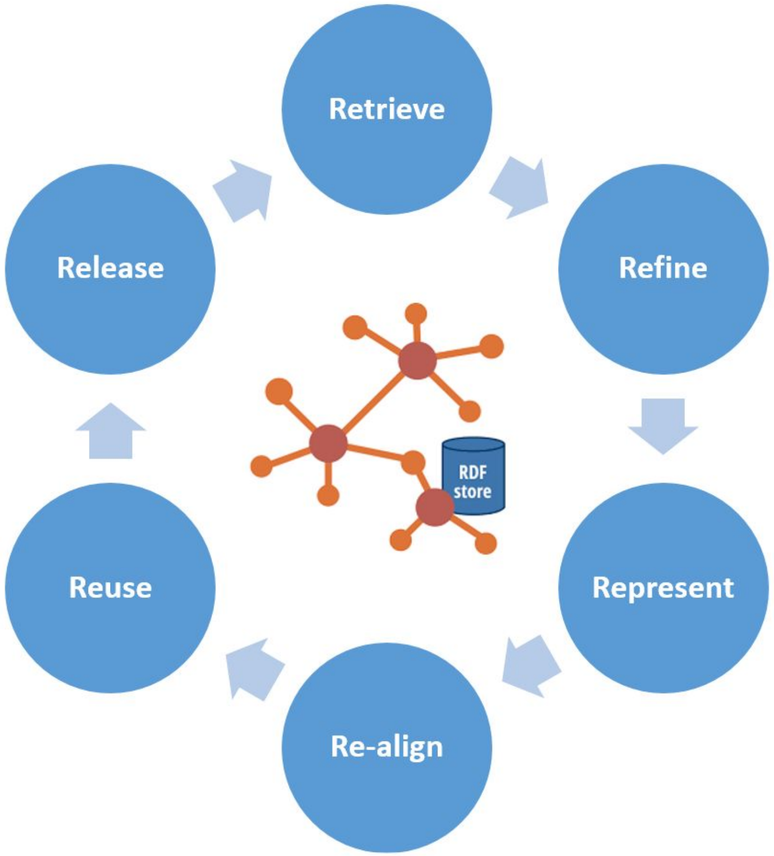
Uncertainty and lack of accuracy: Due the inconsistent data sources, when Ontology learning attempts to define Ontology structures, there is a level of uncertainty in terms of the intent and representation of entities and attributes.
This results in a lower level of accuracy and requires human intervention from domain experts for realignment.

Scalability: One of the primary sources for Ontology learning is the internet, which is an ever growing knowledge repository. The internet is also an unstructured data source for the most part and this makes it difficult to scale the Ontology learning process to cover the width of the domain from large text extracts. One of the ways to address scalability is to leverage new, open source, distributed computing frameworks (such as Hadoop).

Need for post-processing: While Ontology learning is intended to be an automated process, in order to overcome quality issues, we require a level of post-processing. This process need to be planned and governed in detail in order to optimize the speed and accuracy of new Ontology definitions.

15. Ontology learning, Ontology learning process.

The Ontology learning process consists of six Rs:



Retrieve: The knowledge assets are retrieved from the web and application sources from the domain specific stores using web crawls and protocol-based application access. The domain specific terms and axioms are extracted with a calculation of TF/IDF values and by the application of the C-Value / NC Value methods. Commonly used clustering techniques are utilized and the statistical similarity measures are applied on the extracted textual representations of the knowledge assets.

Refine: The assets are cleansed and pruned to improve signal to noise ratio. Here, an algorithmic approach is taken for refinement. In the refinement step, the terms are grouped corresponding to concepts within the knowledge assets.

Represent: In this step, the Ontology learning system arranges the concepts in a hierarchical structure using the unsupervised clustering method (at this point, understand this as a machine learning approach for the segmentation of the data; we will cover the details of unsupervised learning algorithms in the next chapter).

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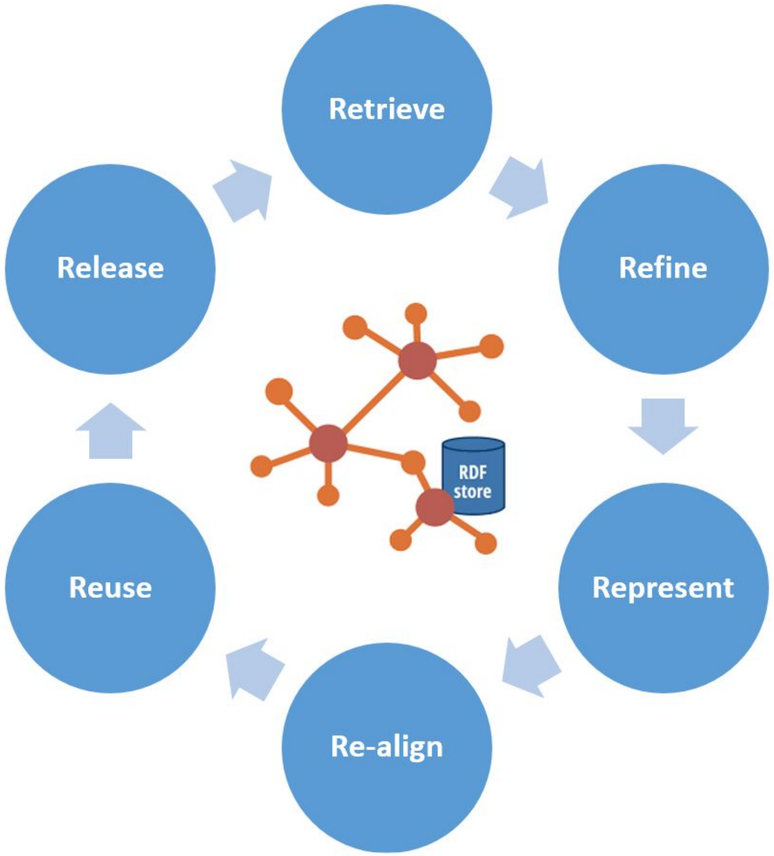

Re-align: This is a type of post-processing step that involves collaboration with the domain experts. At this point, the hierarchies are realigned for accuracy. The Ontologies are aligned with instances of concepts and corresponding attributes along with cardinality constraints (one-to-one, one-to-many, and so on). The rules for defining the syntactic structure are defined in this step.

Reuse: In this step, similar domain-specific Ontologies with connection endpoints are reused, and synonyms are defined in order to avoid parallel representations of the same concepts, which are finalized across other Ontology definitions.

Release: In this step, the Ontologies are released for generic use and further evolution.

Thank you!